Space for rough work

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

10. Arrange the following milestones in the depiction of the structure of an atom in chronological order.

(A) Discovery of nucleus (B) Establishing stability of atom
(C) Discovery of protons (D) Discovery of electrons

(a) CABD (b) DABC (c) CDAB (d) DACB

11. The electronic configurations and atomic numbers of some species are given below. Arrange them in the order of noble gas, cation, non-metal atom, anion and metal atom.

(A) A $\rightarrow$ 2, 8; Z = 11 (B) B $\rightarrow$ 2, 8; Z = 10
(C) C $\rightarrow$ 2, 8, 8; Z = 16 (D) D $\rightarrow$ 2, 8, 2; Z = 12
(E) E $\rightarrow$ 2, 8, 5; Z = 15

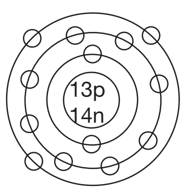
(a) AECDB (b) BCEAD
(c) ADECB (d) BAECD

12. Match the values of Column A with those of Column B.

Column A (Atomic Number)	Column B (No. of Electrons in Penultimate Shell)
(i) 36	(A) 2
(ii) 17	(B) 18
(iii) 25	(C) 10
(iv) 9	(D) 8
(v) 22	(E) 13

(a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (E); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A); (v) $\rightarrow$ (D)
(b) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (E); (iv) $\rightarrow$ (A); (v) $\rightarrow$ (C)
(c) (i) $\rightarrow$ (E); (ii) $\rightarrow$ (B); (iii) $\rightarrow$ (C); (iv) $\rightarrow$ (D); (v) $\rightarrow$ (A)
(d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C); (v) $\rightarrow$ (E)

13. Observe the geometric representation given below and identify the wrong statement.



(a) The species can form a trinegative ion.

(b) The ion resulting from it gets neon configuration.

(c) The mass number of the atom is 27.

(d) The atomic number of the atom is 13.

14. The maximum number of electrons in an orbit is 50. Identify the orbit corresponding to the above number of electrons.

(a) 4 (b) 6 (c) 5 (d) 10

15. The neutral atom corresponding to a positive ion of an element has 2 incomplete shells. The sum of the electrons in the valence shell and the penultimate shell is twice the number of electrons in the antepenultimate shell. Calculate the atomic number of the element.

(a) 26 (b) 28 (c) 30 (d) 24

Space for rough work

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Arrange the following elements in the increasing order of the number of neutrons.

   (A) $_{58}Ce^{140}$ (B) $_{92}U^{238}$
   (C) $_{79}Au^{197}$ (D) $_{90}Th^{232}$
   (E) $_{78}Pt^{195}$

   (a) BECDA (b) BDCEA
   (c) AECDB (d) ADCEB

2. Ions named as 'A', 'B', 'C', 'D' are given below. All of them contain 10 electrons each. Arrange the ions in the increasing order of the atomic numbers.

   (A) A^{+} (B) B^{+2} (C) C^{-} (D) D^{-2}

   (a) DCAB (b) DACB (c) BCAD (d) BACD

3. **Assertion (A):** Eight electrons are present in the antepenultimate shell of Al.

   Reason (R): The electronic arrangement of Al is 2, 8, 3

   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

4. **Assertion(A):** Atoms of the same element possess the same number of total fundamental particles.

   Reason(R): Atomic number is the characteristic of an element.

   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

5. Mass of total positive charge present in an atom is 23,881 times to that of the mass of an electron. Identify the atomic number.

   (a) 6 (b) 12 (c) 13 (d) 14

6. If the mass of a proton is m, what should be the approximate mass of an α-particle?

   (a) m (b) $2m$ (c) $4m$ (d) $6m$

Space for rough work

7. Identify the drawback of Thomson's atomic model.
   (a) Presence of electrons
   (b) Cancellation of positive and negative charges in order to maintain the neutrality of the atom
   (c) Coexistence of opposite charges in the atom
   (d) Both (b) and (c)

8. An element 'X' has the same number of protons, electrons and neutrons, and its atomic mass is double the atomic number. If the number of neutrons is 20, then the element 'X' is
   (a) Ca (b) Mg (c) Zn (d) Mg

9. Which of the following atomic numbers corresponds to the formation of tripositive ion?
   (a) 10 (b) 11 (c) 12 (d) 13

10. An atom of an element has the electronic configuration 2, 8, 18, and 8. Identify the false statement regarding this element.
   (a) The atom has one incomplete shell.
   (b) The atom cannot form any ion.
   (c) The next electron enters the fourth shell.
   (d) The given element is krypton.

11. An atom A of an element with atomic number Z has the electronic configuration a, $a + b$, $2b$, and a. Which of the following statements is false regarding the element?
   (a) The element has 2 incomplete shells.
   (b) The element can have more electrons in its penultimate shell.
   (c) The element with Z + 6 atomic number has 3 electrons in its valence shell.
   (d) Atomic number of the atom is 24.

12. The velocity of an electron decreases with an increase in the distance from the nucleus. Identify the reason.
   (a) Increase in nuclear force of attraction
   (b) Decrease in nuclear force of attraction
   (c) Increase in kinetic energy of electron
   (d) Increase in radius of the orbit

13. Which postulate of Rutherford's theory was not based on the experimental results of α-ray scattering experiment?
   (a) Presence of small central nucleus
   (b) Presence of electrons outside the nucleus

(c) Revolution of electrons around the nucleus
(d) Presence of large empty space in the atom

14. Among the following sets of elements, which set contains the same number of valence electrons?

(a) ${}_{12}A^{24}$, ${}_{19}B^{39}$
(b) ${}_{14}A^{28}$, ${}_{6}B^{12}$
(c) ${}_{12}A^{24}$, ${}_{14}B^{28}$
(d) ${}_{15}A^{30}$, ${}_{8}B^{16}$

15. Match the entries of Column A with those of Column B.

Column A (Element)	Column B (Number of Core Electrons)
(i) Potassium	(A) 2
(ii) Carbon	(B) 10
(iii) Sulphur	(C) 18
(iv) Titanium	(D) 20
	(E) 28

(a) (i) → (D); (ii) → (C); (iii) → (B); (iv) → (A)
(b) (i) → (C); (ii) → (A); (iii) → (B); (iv) → (D)
(c) (i) → (C); (ii) → (D); (iii) → (A); (iv) → (B)
(d) (i) → (B); (ii) → (A); (iii) → (C); (iv) → (D)

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. The atomic numbers of some elements are given below. Arrange the elements in the decreasing order of the number of valence electrons.
   (A) A $\rightarrow$ 16 (B) B $\rightarrow$ 20
   (C) C $\rightarrow$ 14 (D) D $\rightarrow$ 17
   (E) E $\rightarrow$ 13
   (a) DECAB (b) BECAD
   (c) BACED (d) DACEB

2. The number of electrons and neutrons in atoms of different elements are given below. Arrange them in the decreasing order of their mass numbers.
   (A) 38, 49 (B) 56, 81
   (C) 76,114 (D) 47, 61
   (E) 44, 57
   (a) CEDBA (b) CBDEA
   (c) ABDEC (d) AEDBC

3. **Assertion (A):** $_1H^1$ and $_1H^2$ are the isotopes of hydrogen.
   Reason (R): Isotopes have the same mass number and different atomic number.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

4. **Assertion (A):** Atoms are divisible.
   Reason (R): Atoms were found to contain fundamental particles.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

5. The mass of total positively charged particles present in a dinegative ion of an element is 29,392 times that of an electron. Identify the element among the following.
   (a) $_8O^{16}$ (b) $_{16}S^{32}$ (c) $_{14}Si^{28}$ (d) $_9F^{18}$

Space for rough work

6. How many fundamental particles are present in an α-particle?
(a) 2 (b) 4 (c) 6 (d) 3

7. Name the scientist who first established the presence of nucleus in an atom.
(a) J. J. Thomson (b) Rutherford
(c) Bohr (d) Dalton

8. Which of the following metals can be considered the best substitute of gold in α-ray scattering experiment?
(a) Aluminium (b) Iron
(c) Platinum (d) Copper

9. Which of the following atomic numbers corresponds to the formation of trinegative ion?
(a) 7 (b) 15
(c) 18 (d) Both (a) and (b)

10. The number of valence electrons present in noble gases is _____.
(a) 6 (b) 18 (c) 8 (d) 4

11. An atom of an element 'X' has the electronic configuration a, $a + b$, $2(b - 1)$, and a. Identify the atomic number of the element.
(a) 20 (b) 22 (c) 24 (d) 26

12. Identify the false statement according to Bohr's theory.
(a) Electron absorbs energy when it moves from M shell to L shell.
(b) Electron releases energy when it jumps from M shell to K shell.
(c) Electron revolving in K shell does not lose energy at all.
(d) Energy of electron in a given orbit is constant.

13. What was the expected observation of α-ray scattering experiment if Thomson's atomic model would be correct?
(a) All the α-particles would be completely rebounded.
(b) All the α-particles would be deflected at very large angles.
(c) Absence of scintillations on ZnS screen.
(d) All the α-particles would pass straight through the gold foil.

14. Total number of electrons present in the antepenultimate shell of an atom of an element with atomic number 24 is _____.
(a) 2 (b) 8 (c) 13 (d) 1

15. Match the entries of Column A with those of Column B.

Column A (Element)	Column B (Number of Electrons in Penultimate Shell)
(i) Chlorine	(A) 9
(ii) Oxygen	(B) 18
(iii) Zinc	(C) 2
(iv) Scandium	(D) 8

(a) (i) → (B); (ii) → (C); (iii) → (D); (iv) → (A)
(b) (i) → (B); (ii) → (A); (iii) → (D); (iv) → (C)
(c) (i) → (B); (ii) → (D); (iii) → (A); (iv) → (C)
(d) (i) → (D); (ii) → (C); (iii) → (B); (iv) → (A)

Space for rough work

Answer Keys

Assessment Test I

1. (a) **2.** (d) **3.** (c) **4.** (c) **5.** (d) **6.** (b) **7.** (c) **8.** (b) **9.** (d) **10.** (c)
11. (a) **12.** (a) **13.** (c) **14.** (b) **15.** (a)

Assessment Test II

1. (c) **2.** (d) **3.** (c) **4.** (b) **5.** (d) **6.** (c) **7.** (b) **8.** (c) **9.** (a) **10.** (b)
11. (d) **12.** (b) **13.** (a) **14.** (c) **15.** (a)

Assessment Test III

1. (c) **2.** (a) **3.** (d) **4.** (d) **5.** (c) **6.** (c) **7.** (c) **8.** (a) **9.** (d) **10.** (c)
11. (c) **12.** (b) **13.** (c) **14.** (b) **15.** (b)

Assessment Test IV

1. (d) **2.** (d) **3.** (c) **4.** (a) **5.** (b) **6.** (b) **7.** (a) **8.** (c) **9.** (d) **10.** (c)
11. (b) **12.** (a) **13.** (d) **14.** (b) **15.** (d)

Classification of Matter 2

Reference: Coursebook - IIT Foundation Chemistry Class 8; Chapter - Classification of Matter; Page number - 2.1–2.19

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. Some amount of water is kept in a vacuum chamber. In which state will the water exist in the chamber?
   (a) Solid (b) Liquid (c) Vapour (d) Gas

2. When we exhale air from our mouth on a winter morning, it appears foggy. Which of the following interconversions of states of matter takes place here?
   (a) Melting (b) Evaporation
   (c) Condensation (d) Solidification

3. Which of the following statements is correct?
   (a) Diamond is a very good conductor of heat because of the compact arrangement of small carbon atoms.
   (b) Copper is used as telecommunication cable because of its high density.
   (c) Aluminium is widely used in the manufacture of electric transmission cable only because of its light weight.
   (d) Phosphorus finds application in the match industry because of its resistance to heat.

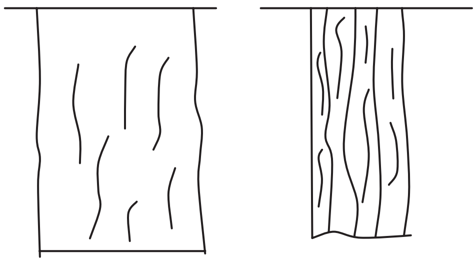
4. Which of the following is not a homogeneous mixture?
   (a) Liquor ammonia (b) Hydrochloric acid
   (c) Soda water (d) Smoke

5. Find the odd one out with respect to hardness.
   (a) Iron (b) Copper
   (c) Aluminium (d) Sodium

6. Alcohol is mixed with sugar solution and benzene is then added to this mixture. Which of the following is the correct sequence of methods to separate this mixture?
   (a) Evaporation, fractional distillation and separating funnel
   (b) Separating funnel, fractional distillation and evaporation

(c) Evaporation, separating funnel and fractional distillation
(d) Fractional distillation, separating funnel and evaporation

7. The same fabric is kept for drying in two different ways as shown in the figures given below. In position A, the cloth dries up faster because the rate of evaporation is _______.



Position A (Day)   Position B (Night)

(a) directly proportional to temperature
(b) directly proportional to surface area
(c) inversely proportional to humidity
(d) Both (a) and (b).

8. A and B are two gases taken in two different cylinders at 45°C. Their critical temperatures are 40°C and 50°C, respectively. Which of the following statements is correct?
(a) Only gas A can be liquefied by applying pressure at that temperature.
(b) Only gas B can be liquefied by applying pressure at that temperature.
(c) Both the gases can be liquefied by applying pressure at that temperature.
(d) None of the gases can be liquefied by applying pressure at that temperature.

9. From the description given below, identify the processes in which the potential energy decreases.
A: Formation of ice
B: Melting of ice
C: Condensation of water vapour
D: Boiling of water
(a) A and B   (b) A and C
(c) B and D   (d) C and D

10. The boiling point of water is directly proportional to the external pressure. Which of the following statements supports the given statement?
(a) Addition of a pinch of salt to water makes the cooking of food faster.
(b) Pressure cooker decreases the cooking time.
(c) Sea water takes more time to be boiled than same mass of normal tap water, if the rate of supply of heat is same.
(d) The same amount of water kept in a wide container and in a narrow vessel evaporates at different rates.